

From Pong to Balatro: What Fifty Years of Great Games Teach Us About the Foundations of Fun

TL;DR

- Across five decades and wildly different forms — a two-paddle arcade machine, a falling-blocks puzzle, a 350-million-selling sandbox, an \$8-billion AR phenomenon, and a one-person poker roguelike — the games that endure share a small set of *philosophical* foundations, not technical ones: they are **easy to learn and hard to master**, built on a **tight core loop of action → feedback → reward**, full of **meaningful choices**, and tuned to keep players in a **flow state** between boredom and anxiety.
- Commercial scale and cultural longevity track these design fundamentals more closely than budget or graphics: Minecraft (350M+ copies), Tetris (520M+ across versions), GTA V (~225M units and ~\$8.5B revenue), and single-developer hits like Stardew Valley (41M+) and Balatro all obey the same principle Nolan Bushnell articulated for Pong in 1971.
- The deepest common denominator is that **a great game is a machine for learning** — Raph Koster's thesis that "fun is just another word for learning." Whether the pattern being mastered is a block stack, a combat-timing window, a social bluff, or a build order, great games deliver a steady stream of solvable problems with instant feedback, escalating difficulty, and emotional or social stakes that make the mastery matter.

Key Findings

1. "Easy to learn, hard to master" is the closest thing the medium has to a universal law. Coined as *Bushnell's Law* by Atari founder Nolan Bushnell around 1971 ("All the best games are easy to learn and difficult to master. They should reward the first quarter and the hundredth"), [Wikipedia](#) it was born from the *failure* of his complex first game, Computer Space — whose four equivalent-looking buttons, per game scholar Ian Bogost, "had confused players... due to their lack of familiarity" — and the success of the one-sentence-teachable Pong. Blizzard adopted it as a formal motto for Diablo, Warcraft, and World of Warcraft; [Wolfshead Online](#) it explains Tetris, Flappy Bird, Minecraft, League of Legends, and Elden Ring equally well.

2. Great games are “machines for learning.” Designer Raph Koster’s *A Theory of Fun* (2005) argues games are “exceptionally tasty patterns to eat up” and that “fun is just another word for learning” — the brain releases endorphins/dopamine when it masters a pattern. Crucially, Koster’s corollary defines both failure modes: a game gets boring when the pattern is fully learned (mastery achieved), OR when it’s too trivial, OR too hard to even perceive. The definition of a good game becomes “one that teaches everything it has to offer before the player stops playing.” [Theoryoffun](#)

3. Flow is the target mental state, and it has a measurable shape. Psychologist Mihaly Csikszentmihalyi’s flow theory — full immersion, energized focus, loss of time-sense — occurs in a narrow channel where challenge matches skill. Too much challenge → anxiety; too little → boredom. Every well-tuned difficulty curve (Mario’s escalating worlds, Celeste’s chapters, a roguelike’s scaling runs) is an engineering of this channel. Flow’s preconditions — clear goals, immediate feedback, challenge-skill balance — are themselves a design checklist.

4. A game is “a series of interesting decisions.” Sid Meier’s famous aphorism frames the other half of engagement: a decision is interesting only when no option is obviously best, options aren’t equally attractive, and the player has enough information to choose meaningfully. This is the engine of Civilization, but also of every loadout choice in a shooter, every joker kept in Balatro, and every “fight or flee” in Dark Souls.

5. The core/compulsion loop is the atomic unit of engagement. The repeating cycle — perform an action, get a reward, unlock a new possibility, repeat — is what keeps a player in the chair. Its psychological roots run to B.F. Skinner’s operant conditioning and variable-ratio reinforcement; its canonical game-design articulation is John Hopson’s “Behavioral Game Design” (Gamasutra, 2001). The same loop underlies Candy Crush’s match-three, WoW’s quest-reward-level cycle, and Pokémon’s catch-train-battle.

6. The most influential designers all preach “fun/idea first, technology later.” Shigeru Miyamoto (gameplay before graphics or story; “the player is the hero, not the designer”; the famous World 1-1 Goomba tutorial), Will Wright (“I’m giving the player a toy, and the player is turning it into a game”; [MasterClass](#) “possibility spaces”), and Sid Meier (“find the fun first and then use the technology to enhance the fun”) [A-Z Quotes](#) converge on the same hierarchy of values.

Details

The historical arc, era by era

Arcade era (1972–1983): the laws are set. Pong (1972) proved a game teachable in one sentence (“Avoid missing ball for high score”) could be a phenomenon. Space

Invaders (1978) introduced the high-score table and competitive replay, grossing an estimated \$2 billion by 1982 [Leon](#) and selling ~360,000 cabinets. [Leon](#) Pac-Man (1980) deliberately broadened the audience beyond young men — designer Toru Iwatani built a non-violent eating game to attract women and couples [Wikipedia](#) — becoming the first true gaming mascot, installed on a Guinness-record ~293,822 cabinets and pulling roughly \$6 billion in quarters in 1982 alone, more than US movie theaters and Las Vegas casinos combined that year. [Fandom](#) Donkey Kong (1981) introduced Miyamoto and the platformer. The takeaway: simple controls, instant feedback, escalating difficulty, score-as-motivation.

Console golden age (1983–1995): depth and worlds. Super Mario Bros. (1985) is a masterclass in implicit teaching — Miyamoto and Tezuka designed World 1-1 so the first enemy (a slow Goomba) and the first mushroom teach mechanics without text.

[HotHardware](#) The Legend of Zelda introduced open exploration and the “miniature garden” (*hakoniwa*) philosophy. Tetris (1988; ~520M across versions) is the purest distillation of the learning loop: each falling block is simultaneously a problem and its own solution. [Mind Hacks](#) Psychologists tie its grip to the **Zeigarnik effect** (uncompleted tasks dominate memory) [Medium](#) and the **Tetris effect** (the pattern intrudes into perception after long play) [Wikipedia](#) — evidence that great games rewire cognition. Sonic and Sega’s rivalry pushed the medium’s identity-building.

The franchise era (1989–2013): identity, narrative, and systems at scale.

Pokémon (catch-train-battle loop plus collection and social trading), Final Fantasy (cinematic narrative RPGs), Grand Theft Auto (the open-world sandbox), The Sims (Wright’s “software toy” with no win condition — the best-selling PC franchise of its time), [ResearchGate](#) Halo, and Call of Duty established that franchises sustain themselves by pairing a reliable core loop with novelty in each installment. GTA V (2013) became the second-best-selling game ever — Take-Two’s fiscal Q3 2026 remarks (quarter ended Dec 31, 2025) state it has “sold-in over 225 million units” — and one of the most profitable entertainment products in history, with AFP reporting in December 2023 that it had “brought in more than \$8.5 billion since release.”

PC gaming: depth, mastery, and community. Doom (1993) defined the FPS and modding/user-generated content. StarCraft set the bar for asymmetric balance and birthed esports. World of Warcraft (2004) reached its peak when Blizzard’s October 7, 2010 press release announced its “subscriber base... now exceeds 12 million players worldwide” (reached after the mainland Chinese launch of *Wrath of the Lich King*), generating roughly \$9 billion lifetime and proving that social bonds plus a treadmill of meaningful progression create gaming’s stickiest loops. Minecraft (2011) — confirmed by Mojang and Guinness World Records (April 2025) at over 350 million units, the first videogame ever to pass that mark — is Wright’s “possibility space” thesis realized: simple block rules generating infinite player creativity.

Mobile (2008–present): accessibility as the whole game. Angry Birds (2+ billion downloads), [KTALnews](#) Flappy Bird (one-tap, brutally hard, pulled by its own creator), Candy Crush Saga (5+ billion downloads, [Kevuru Games](#) \$20+ billion lifetime revenue — yet ~96% of players never pay), [Kevuru Games](#) and Pokémon GO (per Sensor Tower it “topped \$6 billion in lifetime player spending” by June 2022, averaging ~\$1 billion/year, and ~\$8 billion by 2024 per AppMagic; 232M peak players in 2016) [Owebest](#) extended Bushnell’s Law to its endpoint: the one-sentence-teachable game on a device in everyone’s pocket, monetized via freemium loops.

Indie renaissance: authorship and emotional resonance. Stardew Valley (one developer, Eric Barone; 41M+ copies), [Comicbook](#) Undertale (Toby Fox; [24/7 Wall St.](#) subverts combat morality, multiple endings), Hollow Knight (Team Cherry; tiny budget, beloved Metroidvania), [Games](#) Celeste (precision platforming as a metaphor for mental health, with generous assist modes), Among Us (a 2018 sleeper that — per SuperData/Nielsen — reached “roughly half a billion monthly active users” in November 2020, “by far the most popular game ever in terms of monthly players,” with the ~\$5 PC version just 3% of players yet 64% of revenue), Hades (Supergiant; a roguelike whose death-loop is the narrative; 10M+ copies), and Balatro (2024; a single developer, “localthunk,” whose poker-rogue-like topped mobile charts above Minecraft). Indies prove the foundations are budget-independent.

Modern AAA and live service. Fortnite (650M+ registered, \$5.7B in 2024, ~\$40B lifetime), [Quantumrun](#) League of Legends (~130M MAU, ~\$19B lifetime), [Quantumrun](#) Elden Ring (FromSoftware’s open-world synthesis, ~17.7M+ on Steam alone), [Gamalytic](#) and Baldur’s Gate 3 (Larian; the first game to sweep all five major Game of the Year awards — Golden Joysticks, BAFTA, GDC, DICE, and The Game Awards) [GamesRadar+](#) [ResetEra](#) anchor the high end. Breath of the Wild (2017) re-grounded open-world design in **multiplicative gameplay**: a “chemistry engine” of simple, consistent rules (fire, water, ice, wind, electricity interacting predictably) [Zelda Universe](#) that lets players invent their own solutions rather than follow fixed ones. [Zelda Universe](#)

The common denominators — synthesized

Stripping away genre, budget, and decade, the games above share these foundations:

1. **Easy to learn, hard to master (Bushnell’s Law).** A low barrier to entry plus a high skill ceiling. Flappy Bird and Go both satisfy it; so do Tetris and Elden Ring.
2. **Learning as the engine of fun (Koster).** Every great game is a stream of solvable patterns. When learning stops, fun stops — which is why endless variation (procedural runs, new levels, PvP opponents, seasonal content) extends lifespan.
3. **The tight core loop with immediate feedback.** Action → feedback → reward →

new possibility. Instant, “juicy” feedback (Meier: always “acknowledge that the player gave some input”) makes the loop satisfying. The compulsion-loop literature (Hopson; Skinner) explains why variable rewards intensify it — and why this power carries ethical weight.

4. **Meaningful choice and player agency (Meier, Wright).** Interesting decisions and large “possibility spaces” make players feel ownership. Wright: players care most about outcomes that are uniquely theirs.
5. **Flow (Csikszentmihalyi).** A difficulty curve that tracks rising skill, with clear goals and feedback, keeps players in the immersive channel between boredom and anxiety.
6. **Emotional resonance and meaning.** Miyamoto’s *kyokan* (shared feeling); Undertale’s moral weight, Celeste’s empathy, Hades’s narrative-through-repetition, BG3’s character-driven choice. The mechanics carry feeling.
7. **Social and community dynamics.** From the arcade crowd around a cabinet, to WoW guilds, to Among Us lobbies, to League’s esports, multiplayer and community convert a good loop into a lasting world. Among Us shows social context can be the decisive multiplier.
8. **Novelty balanced with familiarity.** Bogost’s “catchiness” point: familiar enough to grasp instantly, novel enough to intrigue. Franchises and roguelikes both live in this tension.
9. **Accessibility paired with depth.** Pac-Man’s deliberate broadening, Celeste’s assist modes, Mario’s simple-jump-with-hidden-depth. The best games let beginners and experts find their level in one system.
10. **Emergence over prescription.** Breath of the Wild’s chemistry engine, Minecraft’s blocks, The Sims’ systems — simple consistent rules that multiply into player-authored stories [Zelda Universe](#) outlast hand-scripted, single-solution content.

What Flappy Bird shares with Zelda; what Tetris shares with WoW

Flappy Bird and Breath of the Wild look like opposites — one a one-tap punisher, the other a 60-hour open world. But both are *easy to learn, hard to master*, both give instant unambiguous feedback on every action, and both keep the player in flow by matching challenge to growing skill. Tetris and World of Warcraft seem even further apart, yet both run on the same loop: a never-ending stream of clearly-defined, immediately-actionable tasks (clear a line / complete a quest), each delivering a small reward and a slightly harder next problem — Zeigarnik’s “uncompleted task” pull operating at the scale of seconds (Tetris) or weeks (WoW raid progression). The substrate differs; the

cognitive machine is identical.

Recommendations

For anyone designing, evaluating, or studying games, the synthesis suggests a staged checklist:

1. **Start with the core loop, not the content.** Before art, story, or technology, prototype the 10-to-60-second cycle of action-feedback-reward and confirm it is fun *bare*. Miyamoto spent months making Mario “fun just to move.” Benchmark: if the loop isn’t compelling with placeholder graphics, no polish will save it.
2. **Pass the one-sentence test, then build a skill ceiling.** Can you teach the game in a sentence (Bushnell)? If not, simplify the entry. Then ask whether mastery can deepen for hundreds of hours. Course-correction signal: if testers plateau in interest quickly, the learning-pattern is exhausting too soon (Koster’s failure mode) — add variation or depth.
3. **Engineer the flow channel explicitly.** Map the difficulty curve against expected skill growth; use the “rising sawtooth” (ramp up, brief relief to feel heroic, ramp again). Benchmark: rage-quits signal anxiety (too hard); disengaged drop-off signals boredom (too easy).
4. **Audit every choice for “interestingness” (Meier).** Remove decisions where one option dominates or all are equivalent; ensure players have the information to choose. If a choice isn’t interesting, cut it. Game Developer
5. **Decide where you sit on accessibility vs. depth — and serve both if possible.** Assist modes (Celeste), scalable difficulty (Mario), broadened themes (Pac-Man) expand the audience without diluting mastery — increasingly a commercial as well as ethical expectation.
6. **Use compulsion-loop psychology responsibly.** Variable rewards are powerful and now legally scrutinized (loot boxes, freemium targeting). Recall ~96% of Candy Crush players never pay, Kevuru Games so revenue concentrates on a vulnerable few — treat retention mechanics as a duty of care, not just a KPI.
7. **Plan for emergence and community.** Where feasible, favor simple consistent systems that multiply (chemistry engines, sandboxes) over single-solution content, and design for sharing, watching, and playing together — the multipliers behind Minecraft, Among Us, and League.

Caveats

- **Sales and player figures are noisy.** “Best-selling ever” is genuinely disputed: Minecraft (~350M as a single product) vs. Tetris (~520M only if all versions are aggregated). Take-Two reports GTA V “sold-in” (shipped) figures, not sold-through. Free-to-play “players/installs” (Fortnite, Pokémon GO) are not comparable to paid “copies sold,” and revenue estimates from third parties (Sensor Tower, AppMagic, SuperData) are modeled, not audited — note the Pokémon GO lifetime figure ranges from ~\$6B (Sensor Tower, 2022) to ~\$8B (AppMagic, 2024).
- **The “\$8.5 billion / most profitable entertainment product” claim for GTA V** traces to MarketWatch (2018, originally ~\$6B) and AFP/Barron’s (late 2023, ~\$8.5B); rounder “\$8 billion as of 2025” phrasing circulates mostly in lower-tier outlets recycling those figures.
- **Quote attribution is imperfect.** The Miyamoto line “a delayed game is eventually good...” is likely misattributed (per researcher Kate Willaert). Meier himself has said “a series of interesting decisions” was a casual GDC definition, “not meant to be an all-inclusive... definition.”
- **These are principles, not formulas.** They describe common denominators of success after the fact; they do not guarantee it. Timing, distribution, marketing, and luck (Among Us’s pandemic timing; streamer amplification) are real and sometimes decisive. Some beloved or important games deliberately violate a principle (e.g., punishing difficulty over accessibility) in service of artistic intent.
- **“Fun” is not the only goal.** Critics such as Ian Bogost note Koster’s fun-centric model undervalues games as expression, art, or argument. Narrative and “walking-sim” experiences can matter without optimizing the learning loop.